BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000577	**Timestamp:**	2026-09-08 19:58:57
Input Source:	Upload	**Evaluation Status:**	**NORMAL**
Model Name:	Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%)	**Confidence:**	**94.53%**
Advisory Status:	**NEEDS CONFIRMATION** (τ=70%)	**Advisory Reason:**	**RH_UNCERTAINTY**

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP

Predicted Blood Group:

O-

Confidence Score: **94.53%**
Advisory Threshold: **70%**
Advisory Screening: **NEEDS CONFIRMATION**
Derived ABO Group: **O** (95.7% prob) | Rh Factor: **-**

RESEARCH ADVISORY: Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (predicted Rh-negative requires serological validation due to weak ridge-Rh correlation). ABO group appears to be 'O' (95.7% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
0.0%	0.8%	0.0%	3.4%	0.0%	0.0%	1.2%	94.5%